bpm-linearity guide

Wenu-imaging

10 November 2026

Executive summary

For each detector, load calibrations and raw planes once, capture raw evidence, scan-correct, trim, subtract the supplied active master from flats and retain one processed cube. If an input BPM is supplied, log the skip; otherwise create a robust response BPM. Measure linearity ROIs, pairs and direct sites from that same cube, publish FITS/ASDF, then release it. Linearity is measurement only: no physical response correction, shutter solution or gain conversion is applied.

Qualification. Earlier CCD65 saved-product checks passed. The classified-list interface still requires Mac qualification; WFI and full-mosaic review are pending.

Command and help

Run from orchestration with a fresh empty reduced run:

```
mefpp_recipe cal bpm-linearity --config RECIPE.toml --qc false
```

```
mefpp --help
mefpp_recipe cal --help
mefpp_recipe cal bpm-linearity --help
mefpp_recipe cal linearity --help
mefpp_recipe cal bias --help
```

Help is data-free: categories list recipes and pending entries; operation help reports inputs, options, defaults and outputs.

Public option	Meaning / default
`--config FILE`	Required recipe TOML; references project/settings/inputs.
`--qc true|false`	Optional override; CLI > recipe `qc` > true.
`--help`	Context-specific help; successful exit, no processing.

Inputs and directory contract

Project TOML defines raw, calibration, reduced, work and log roots; its containing directory is orchestration. Raw observations and input calibrations are read-only. TOMLs belong in orchestration. Products, work and logs belong below reduced. Recipe `inputs.list` takes the absolute path of an ordered text list. The list directory is configurable.

```
/data/raw/OMEGA.frame1.fits  FLAT,DOME
/data/raw/OMEGA.frame2.fits  FLAT,DOME
```

Each row is path plus `DPR.TYPE`; tabs permit spaces in paths. Comments and blank lines are ignored. Use absolute FITS paths under the project raw directory. Invalid or duplicate inputs fail. Save order, classifications and bindings.

Supply a reduced master bias and optional BPM; dependency-list reduction is deferred. Optional BIAS rows provide diagnostics, never master construction. Flat pairs derive from classification and header exposure times: exactly two members per group within the profile tolerance; missing/ambiguous partners fail. Review calibration, timing tolerance and site before using the WFI example.

Recipe settings and defaults

Templates under `config/recipes/`:

`omegacam-bpm-linearity.example.toml`, `wfi-bpm-linearity.example.toml`. They initially select one detector; extend in detector-catalogue order after reviewing the actual bindings.

Field	Default / requirement
`project`, `settings`, `run`	Required project/native settings; fresh run relative to reduced.
`qc`	true; optional diagnostics enabled.
`assembly.reference_detector`	Required selected detector; among created BPMs if any.
`preparation.detector_toml`	Required detector identity/geometry file.
`preparation.processing_toml`	Required scan/trim/ROI and measurement controls.
`bpm.acquisition`	Required: `quick` or `provisional`.
`bpm.output_mode`	Required: `new` or `combined`.
`bpm.response_low/high`	0.5 / 1.5; finite, $0 <$ low $<$ high.
`bpm.upper_fraction`	Required, $(0, 1]$; example 0.85.
`bpm.minimum_span_fraction`	Required, $(0, 1]$; example 0.15.
`bpm.maximum_pair_difference`	Required, ≥ 0; example 0.02.
`inputs.list`	Absolute path to the classified observation list.
`detectors`	Explicit detector/configuration/HDU/readout bindings.

Each detector declares `master_bias`, `master_hdu`, `master_sha256`, reference ROI IDs, direct `sites` (ROI, active x/y), and pixel/row bounds. `input_bpm` is an explicit calibration path with `bpm_hdu`, or `omitted`. Common `hdu` is required; optional `flat_hdus`/`bias_hdus` arrays override it per input. Raw paths use the raw root; master/BPM paths use its calibration root. Original indices are list-row ordinals. Rectangular ROIs are defined in the processing profile (`xmin/xmax/ymin/ymax`); named direct sites use single-pixel `x/y`. The two are distinct measurements.

Optional `identity` defaults: `identity_key=EXTNAME`, `amplifier_key=-` (unused), bin keys `HIERARCH ESO DET WIN1 BINX/BINY`, `exposure_key=EXPTIME`, `exposure_hdu=0`. Current profiles require 1x1 binning.

Processing TOML explicitly supplies scan switches, sampled scan rectangles, low/high rejection counts, retained rectangles, ROI rectangles, saturation, mask encoding/polarity/reject bits, pair-time tolerance, bias-admission/stack controls and `with_mad`. These have no silent instrument defaults. OCAM examples use saturation 65534 ADU, rejection 10/10, bias range 0–1000 ADU, stack kappa 2, minimum 3 frames, binary good value 1 and exact pair-time matching. These are configured values, not universal detector properties.

`qc=false` skips optional BPM full-frame QC and MAD, retaining requested ROI means, variances, populations, pair/direct-site data, traces and provenance. `qc=true` respects the explicit `with_mad` setting. Disabling MAD does not disable medians required by the robust response algorithm. Unrequested or unavailable statistics are marked unavailable, never filled with zero.

Per-detector processing and mask policy

1. Validate ordered identities and geometry; load the supplied master and optional input BPM once. Capture raw-only evidence during ingestion.
2. Correct each raw flat/bias plane once and retain corrected active images, required raw planes/validity, identities and row traces in one owned detector cube.
3. If a BPM is supplied, skip creation and log `input_bpm_supplied`. Otherwise fit a candidate from borrowed corrected flats and publish its FITS/QC.
4. Measure ROI, pair and direct-site data from that same cube. Release it before proceeding to the next detector; assemble newly created BPMs afterward.

The admitted input mask stays fixed; a generated candidate never feeds back into this run's measurement populations. Explicit omission removes only the current mask, preserving historical master/archive lineage. ROI populations can increase; previously admitted direct values must remain exact and newly admitted sites are reported separately. Whole-file preservation hashing still reads inputs before and after the batch; numerical ingestion/correction alone is read-once.

Correction and measured quantities

For frame f, row y, sorted enabled scan reads $v_{f,y,(k)}$, and rejection counts l, h, the existing arithmetic is

$$O_f(y) = \mathrm{pix}\left[\frac{\sum_{k=l+1}^{n-h} v_{f,y,(k)}}{n-l-h}\right], \tag{1}$$

$$U_f(x,y) = \mathrm{pix}[D_f(x,y) - O_f(y)], \qquad S_f(p) = \mathrm{pix}[U_f(p) - B(p)] \quad \text{(flats only)}. \tag{2}$$

Scan summation/division use double precision; pix converts to native `pixelvalue`. Preserve the separate subtraction roundings. Trim is FITS 1-based inclusive; active row j maps to raw row $j + y_{\min} - 1$. Each raw row uses its own scan reads, without smoothing or fitting. The master is already scan-corrected/active and receives no further scan correction. Bias-role frames have no master subtraction. Active BPMs do not mask electronic scan reads.

For each ROI, the common population $\mathcal{P}$ passes the fixed mask and finite/raw-saturation checks in both flats and, when supplied, both raw reference biases. Each member retains its own exposure time, identity and measurement. For $N = |\mathcal{P}|$:

$$\bar{S}_i = N^{-1} \sum_{p\in\mathcal{P}} S_{ip}, \qquad s_i^2 = \frac{\sum_{p\in\mathcal{P}} (S_{ip} - \bar{S}_i)^2}{N-1}, \tag{3}$$

$$d_p = \mathrm{pix}[S_{1p} - S_{2p}], \qquad s_d^2 = \frac{\sum_{p\in\mathcal{P}} (d_p - \bar{d})^2}{N-1}. \tag{4}$$

Corrected counts must match admitted raw counts. For $N < 2$, sample scatter is unavailable. Save member variances, difference variance and $s_d^2/2$ distinctly. Spatial variance is not automatically a noise weight or conversion-gain estimate. Direct sites retain physical pixel values; ROI means do not replace them in a nonlinear pixel model. Units remain ADU, ADU^2 and seconds.

With raw BIAS evidence, raw-pair diagnostics retain their original stage and populations:

$$\mathrm{RON}_{\mathrm{ADU}} = \sqrt{v(B_1 - B_2)/2}, \tag{5}$$

$$\mathrm{RON}_{\mathrm{MAD}} = \frac{1.4826}{\sqrt{2}} \,\mathrm{median}\,|\Delta B - \mathrm{median}\,\Delta B|, \tag{6}$$

$$g = \frac{\bar{F}_1 + \bar{F}_2 - \bar{B}_1 - \bar{B}_2}{v(F_1 - F_2) - v(B_1 - B_2)} \quad [\text{electrons/ADU}]. \tag{7}$$

Without raw BIAS evidence, RON/gain diagnostics are explicitly unavailable; corrected ROI/pair/site measurements continue on the two-flat population. Here v is ordinary sample variance on the declared raw population; MAD is optional. Gain requires a positive denominator and suitable flat pairs; it is a recorded diagnostic, not a conversion applied to the cube. Corrected pair-difference variance must not be passed through this formula with a second bias subtraction. Shared-master variance/covariance are unavailable; no physical fit weights or uncertainties are invented from spatial scatter.

Robust response BPM

For corrected flat j, the reference level x_j is the existing native median of finite eligible pixels in the union of reference ROIs. Prior-mask selection controls the reference population, not whether a pixel is evaluated for defects. Per-pixel valid/unsaturated samples fit the existing Eclipse robust line:

$$S_{pj} \simeq \alpha_p + \beta_p x_j, \qquad A_p = N_p^{-1} \sum_j |S_{pj} - \alpha_p - \beta_p x_j|, \tag{8}$$

$$R_p = \mathrm{pix}\left[\beta_p / \,\mathrm{mean}_{q\in G}\,\beta_q\right]. \tag{9}$$

G contains valid reference slopes. Preserve the fitter's median conventions, native rounding and flagged least-squares fallback. A_p is mean absolute residual, not MAD. Unavailable fits have separate coverage states.

Flags, suitability and products

Recipe `response_low/high` set the current native float32 thresholds: $R_p \leq$ low sets bit value 4; $R_p \geq$ high sets value 8. The unflagged interval is strictly inside 0.5–1.5 by default. These are *dimensionless relative-response limits*, not electrons/ADU. The input master's original configuration remains unchanged; save the current effective thresholds separately in provenance.

The unsigned-byte registry defines the following meanings; the current response fitter creates only values 4/8. Other tests require their own reviewed criteria.

Value	Meaning	Value	Meaning
1	Bias offset	16	Nonlinear response
2	Dark current	32	Temporal noise
4	Low/dead response	64	Instability
8	Excess response	128	Premature saturation

Bits combine by OR. Zero means no detected reason, not certification. Coverage is separate: 0 unavailable, 1 fitted, 2 flagged fallback. `new` output contains newly measured reasons; `combined` requires a known reason-mask prior. A binary prior cannot supply unknown historical reason bits.

Raw saturation and `upper_fraction` limit usable samples; the upper margin and minimum signal span are different controls. Quick mode requires at least five usable pairs with five distinct times/paired levels; provisional mode requires two levels. Save sample counts, signal coverage and pair mismatch. For positive pair reference levels x_1, x_2, the saved mismatch is $|x_1 - x_2|/[(x_1 + x_2)/2]$; compare it with `maximum_pair_difference`. The shared-cube path records failed suitability and continues measuring; it does not relax thresholds, erase evidence or declare a scientifically accepted BPM.

Product	Contents
Detector BPM FITS	Reason image; `COVERAGE`, `NSAMPLE`, `NEWFLAGS`, `SELECTION`, optional `PRIORSEL`.
Detector ASDF	BPM QC, reason/coverage counts, settings, input/HDU mapping and calibrations; linearity measurement ASDF retains ROI/pair/site data and traces.
Final BPM MEF/ASDF	Only newly created BPMs; none for an all-supplied-mask run.
Logs and `native-recipe.toml`	Stage/skip/failure messages, effective recipe, input preservation and truthful completion. Failed work is retained.

BPM image products use `CMDP.<recipe-start>.001.fits` with matching ASDF/log names; existing destinations are refused. A zero exit, completion receipt and independent saved-product checks are required. Saved readers and viewers use measured ASDF values without raw-pixel access or refitting. Separate exploratory residual analysis and future shutter/linearity model specifications remain in their separate guides linked from `docs/developer/README.md`; their eventual consolidation is deferred. They are not steps in this measurement-only recipe. Physical correction and cross-talk remain deferred.

Source map

Entry point: `mefpp2/ins/mef/recipes/mefpp_bpm_linearity.c`. Helpers below are in `mefpp2/ins/mef/common/`.

Stage	Source / landmark
Admission	`mef_bpm_linearity_compile.c`, `mef_project.c`; config/inputs/bindings helpers; `BPM-LINEARITY-ADMISSION`.
Detector/cube	`mef_bpm_linearity_process.c`, `mef_detector_cube.c`; `BPM-LINEARITY-INGESTION`.
BPM / measurements	`mef_cal_bpm.c`, `mef_linearity_pairs.c`, `mef_linearity_prepare.c`; `BPM-LINEARITY-CANDIDATE`, `BPM-LINEARITY-MEASUREMENTS`.
Publication	`mef_bpm_linearity_products.c`, shared native ASDF writer; `BPM-LINEARITY-PUBLICATION`, `BPM-LINEARITY-PRESERVATION`.

Robust arithmetic: `mefpp2/src/iproc/detector_response.c` and `mefpp2/src/math/fit_curve.c`. Scan arithmetic: `mef_overscan_correction_traced`; statistics: `image_pair_stats_checked`.